

There are a number of sophisticated ways in which analysts could try to adjust returns to reflect risks, many of which rely on multi-factor pricing models. Again, we encourage analysts to avail themselves of these models and to perform these risk-adjustment exercises whenever possible.

But for the purpose of this chapter, we'll focus on one of the simplest approaches possible: we'll simply calculate the return per unit of risk, where risk is taken to be the standard deviation. In other words, we'll divide the return by the risk.

In this chapter, we refer to this measure as the Sharpe ratio. Strictly speaking, the Sharpe ratio is the ratio of excess return to standard deviation, where excess return is defined as the difference between the return of the asset and the risk-free rate over the same period. In this chapter, we take liberties by referring to the ratio of return to risk with the term Sharpe ratio. In most of our analysis, trades already are financed at a rate equal to (or very close to) the risk-free rate. So the returns we experience already are net of the risk-free rate. So we actually are dealing with Sharpe ratios, properly defined. However, we appreciate that this won't be the case for everyone, and we encourage people to be sure to use excess returns rather than unadjusted returns when calculating Sharpe ratios.

Once we have specified an SDE to model the mean reversion of a trade, and once we have calibrated the parameters of the model (most likely, but not necessarily, estimating these via historical data), we can calculate the conditional expected value and the conditional standard deviation of the return distribution.

In the case of simple models, such as the OU process, the transition densities are normal, and the expected values and the standard deviations of the transition densities can be calculated analytically via closed-form expressions.

In the case of more complex models, we face a number of choices, as mentioned above, including numerical solutions to partial differential equations or series approximations to the actual densities.

In our experience, these methods are relatively difficult to implement in practice, and we prefer simply to calculate these conditional moments via simulation. Once the conditional expected value and the conditional standard deviation of the return have been calculated, the conditional Sharpe ratio can be calculated simply by dividing the conditional expected return by the conditional standard deviation.

Conditional Sharpe ratios can be calculated for each individual trade in a portfolio, but we remind readers that the relevant consideration is the effect that inclusion of a trade has on the conditional risk-adjusted return of the entire portfolio. In other words, when evaluating the attractiveness of a trade, the important consideration is not the risk-adjusted return of the trade on a

stand-alone basis but rather the extent to which adding a position can improve the risk-adjusted return of the overall portfolio.¹

We have described above how the general SDE can be interpreted as separating and connecting the predictable return (e.g. from mean reversion) and unpredictable risk of a trade. Using the Excel sheet accompanying this chapter, we will now illustrate how the representation of a trade via such an SDE can be used to estimate its Sharpe ratio. Modeling the trade via an OU process, the sheet “Calc” shows in column L the expected future path of the trade, reflecting the return predictability from mean reversion. Hence, the expected (excess) return after n weeks can be calculated by subtracting the expected value after n weeks (shown in the n -th row below cell L2) from the entry level (shown in cell L2). The modeled evolution of the variance is given in column O. The Sharpe ratio at a specific time in the future, say, 4 weeks after entry, can be calculated by dividing L2 minus L6 by the square root of O6, resulting in a 4-week Sharpe ratio of 0.76 (annualized 2.73, by multiplying with the square root of 52/4).

One can also obtain a path of the expected annualized Sharpe ratio for holding the trade over different horizons. When trades are modeled with an OU process, the expected annualized Sharpe ratio decreases with the holding horizon. This reflects the fact that the contribution from mean reversion is higher, the further the trade is away from its mean; as it approaches its mean, the return predictability from further mean reversion decreases relative to the risk from randomness, which the OU process assumes to be constant.² Consequently, at least when modeling trades with OU, it will usually appear advisable to use short holding horizons and tight targets (e.g. half the distance to the mean rather than the mean itself), as described below.

First Passage Times

Consider a swap spread trading currently at 40 basis points (bp). Let’s imagine that we start a stopwatch with the spread at 40 bp, and then we stop the

¹In most relative value analysis of fixed income markets, we tend to model swap rates and bond yields rather than swap net present values and bond prices. As these tend to be interest-bearing instruments, one needs to consider not only the change in the interest rate over time but also the fact that these instruments pay interest, may involve financing, and experience “pull-to-par” effects over time – a collection of features often referenced under the broad rubric of “carry.” Spread trades with positive carry are particularly attractive in that they provide an additional positive bias to a trade, and these considerations should be included when calculating ex ante Sharpe ratios as well.

²By contrast, if volatility also decreases as the trade approaches its mean, it is conceivable that the expected Sharpe ratio will increase with the holding period.

stopwatch when the spread first reaches a level of 50 bp. The first passage time from 40 bp to 50 bp is the amount of time recorded by the stopwatch when it reaches this 50 bp level. In this case, it's the amount of time it takes for the spread to hit 50 bp, given that it's currently trading at 40 bp. For this reason, the first passage time is also referred to as the *hitting time*, and we'll use these two terms interchangeably.

The first thing to note about the first passage time is that it is specified by two values: a starting value and an ending value. In our example, the starting value was 40 bp, and the ending value was 50 bp. But we could have just as easily specified that the ending value would be 35 bp, or 20 bp. But of course, if we had specified a different stopping value, we would be considering a different hitting time.

The second thing to note about these first passage times is that they're random variables. We don't know the amount of time that will pass before our swap spread hits 50 bp. It could take 10 minutes, or it could take 10 weeks. It could even take 10 months or 10 years.

As random variables, we can calculate moments and other expectations of functions of hitting times. In other words, we can calculate means, standard deviations, skewness, kurtosis, densities, distributions, etc. for first passage times.

As an illustration, imagine that the SDE followed by the swap spread in our example was $dx = k(\mu - x)dt + v dw$, with $k = 0.30$, $\mu = 20$ and $v = 20$. In this case, we can generate, say, 10,000 simulated paths, all starting at 40 bp and then calculate the first passage times to a level of 50 bp along these sample paths. We'll have a simulated sample of 10,000 hitting times, which we can use to make a histogram, generate a nonparametric density, calculate an average hitting time, etc. In particular, Figure 2.16 illustrates a histogram and a nonparametric density resulting from this exercise.

The hitting times have a number of practical applications, in that they allow us to answer a number of very useful questions. For example, consider a few useful questions in the context of trading:

- What is the likelihood of stopping out of a trade during the next two weeks if I set my stop at 55 bp?
- At what level should I set my stop if I want the probability of hitting my stop within the next month to be 20%?
- What is the likelihood of hitting my profit target during the next two months if I set it at 35 bp?
- How much time should I expect to pass before this spread next reaches its long-term mean?
- What is a 70% symmetric confidence interval for the amount of time that will pass before this spread next reaches its long-run mean?

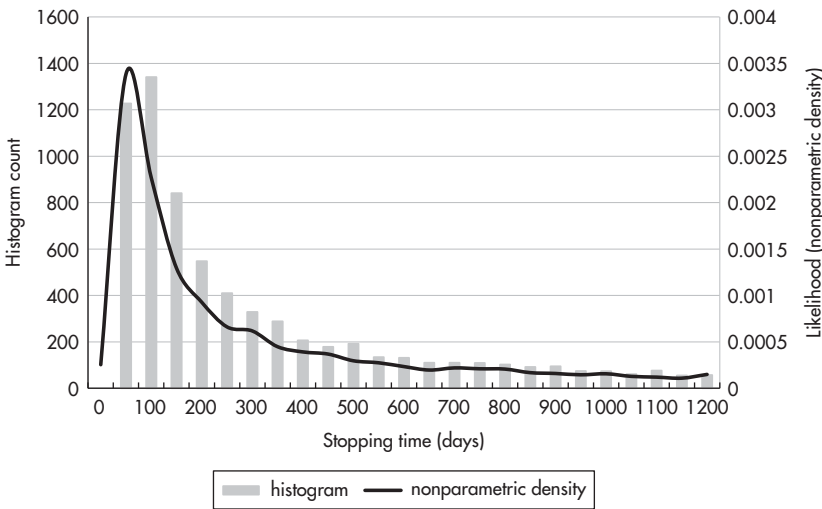


FIGURE 2.16 Nonparametric kernel density and histogram.
Source: Authors.

Note that these last two questions can't be answered simply by considering the transition densities for the process used to model the spread. For example, let's consider a case in which we have used a linear specification for the drift coefficient in the SDE used to model our variable. In that case, the expected value of our variable as a function of time will be exponential, approaching the long-run mean asymptotically but never actually reaching the long-run mean.

In this case, we might ask a different question that gives a false impression of being similar, namely, "How far into the future is the expected value of my variable equal to the long-run mean?" In this case, the answer is that there is no point in the future at which the expected value of the process is equal to the long-run mean, as the expected value approaches the long-run mean asymptotically. In this case, therefore, we might erroneously conclude that expected first passage time to the long-run mean is infinite.

We can see intuitively that this is the wrong answer simply by thinking of a simulation exercise. If we start the process at the current value of our variable and simulate it over a long period, it's very likely that the process will cross back and forth over the long-run mean quite a few times.

The key for anyone who finds the relation between these two questions paradoxical is to appreciate that they're two different questions. One question refers to the path of expected values as a function of time, whereas the other question refers to the expected values of the times taken to travel a certain distance along a sample path.

ASSESSING AND OPTIMIZING EXECUTION STRATEGIES

A major benefit of modeling a trade via an SDE is the ability to assess and optimize execution strategies, for example, the target and stop loss as well as scaling in and out of a trade as a function of its distance from the target. We have found that many traders, while spending much time thinking about *which* position to put on, do not spend much time thinking about *how* to execute it. Rather than aiming for the best execution strategy for a specific position, they may apply general rules-of-thumb to set their target and stop loss. Using an SDE to play around with different targets and stop loss levels can then provide an eye-opening experience: it is not uncommon for the expected Sharpe ratio of the same trade to double simply by using a more appropriate target and stop loss level.

Likewise, if an investor is subject to constraints such as maximum draw-down levels, he can apply an SDE to assess the impact of different ways to fulfill his constraints on the expected performance. For example, he may find that meeting the constraint only by setting a stop loss would result in a too low expected Sharpe ratio, but that adding a scaling strategy increases the anticipated performance to an acceptable level.

While the optimal execution strategy depends on the individual trade, the SDE used to model it and the goals and constraints of the trader, we can share a few general experiences:

- Since the contribution from mean reversion increases with the distance from the mean, trades lose their attractiveness as they approach their mean. Hence, it is usually not advisable to hold a trade until it hits its mean. Setting the target at half the way to the mean or (depending on the alternative trades available) even tighter levels therefore results in higher expected Sharpe ratios.
- Rather than setting the stop loss at a fixed level, it could follow the evolution of the 2-sigma bands of the SDE, for instance (see the chart in the sheet accompanying this chapter).
- Scaling in and out of a position as it moves away and toward its mean is an execution strategy which embodies the idea of mean reversion: The further away from the mean, the higher the expected Sharpe ratio, *hence* the higher the size should be. In line with this, using scaling strategies usually results in at times significantly higher expected Sharpe ratios. However, scaling conceptually conflicts with stop loss: as the trade moves away from its mean and target, the size is increased, until it hits its stop loss, where the size is suddenly set to zero. Hence, while scaling is a suitable execution strategy to increase the return of mean reverting positions, its applicability depends on the individual constraints of the investor.

A PRACTICAL EXAMPLE INCORPORATING ALL THE IDEAS

The Difference between EUR and GBP 5Y5Y Swaption Volatilities

As an example, let's consider the difference between implied volatilities, expressed in bp/year, between 5Y (five-year) options on 5Y swaps denominated in EUR and GBP from 25-May-05 until 4-Jul-12.

The two series are shown in Figure 2.17.

The main point to note from Figure 2.17 is the strong correlation between the two series over time. In particular, the correlation over the period shown in the graph was 0.87.

Figure 2.18 shows the EUR 5Y5Y swaption volatility less the GBP 5Y5Y swaption volatility over the same period.

There are two points worth noting in Figure 2.18.

- The difference between the EUR and GBP swaption volatilities does appear to have been mean reverting over the period.
- The difference appears to be more volatile during the second portion of the data sample than it does during the first portion.

To illustrate this aspect of the data more clearly, Figure 2.19 shows the daily changes in the series displayed in Figure 2.18.

Figure 2.19 supports our intuition from Figure 2.18 that the volatility of the difference has increased during our sample. There are approaches for dealing with changes in parameter values over time, but these are beyond the scope of this book. For our purposes, we'll simply note this feature of the data and keep it in mind as we draw inferences from our analysis.



FIGURE 2.17 EUR and GBP 5Y5Y implied swaption volatilities.

Source: data – Bloomberg, chart – Authors.

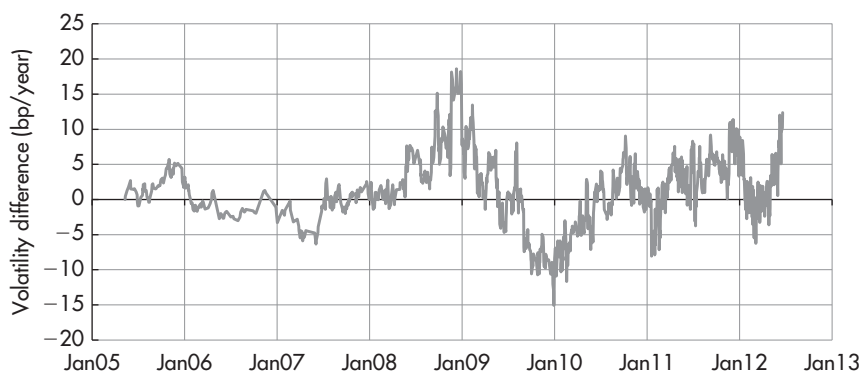


FIGURE 2.18 Swaption volatility difference: EUR 5Y5Y – GBP 5Y5Y.

Source: data – Bloomberg, chart – Authors.

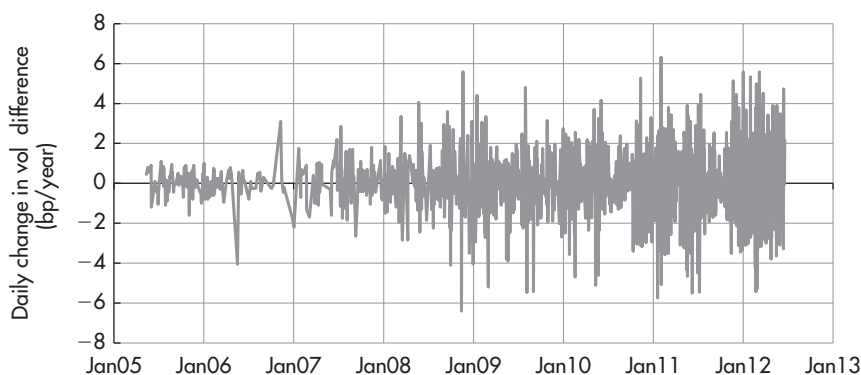


FIGURE 2.19 Daily change in swaption volatility difference series.

Source: data – Bloomberg, chart – Authors.

The next step in the analysis is to consider the diagnostic graphs discussed above. In particular, the first-order nonparametric estimates of the drift and diffusion coefficients are shown in Figure 2.20 and Figure 2.21.

As one would hope when dealing with a mean-reverting process, the average change is negative for values above the mean, and the average change is positive for values below the mean. Otherwise, there are no obvious strong nonlinearities apparent that might cause us to consider using a nonlinear specification for the drift coefficient.

In this case, we may wish to investigate further the benefits of a nonlinear specification for the diffusion coefficient, given that a bucket at either end of

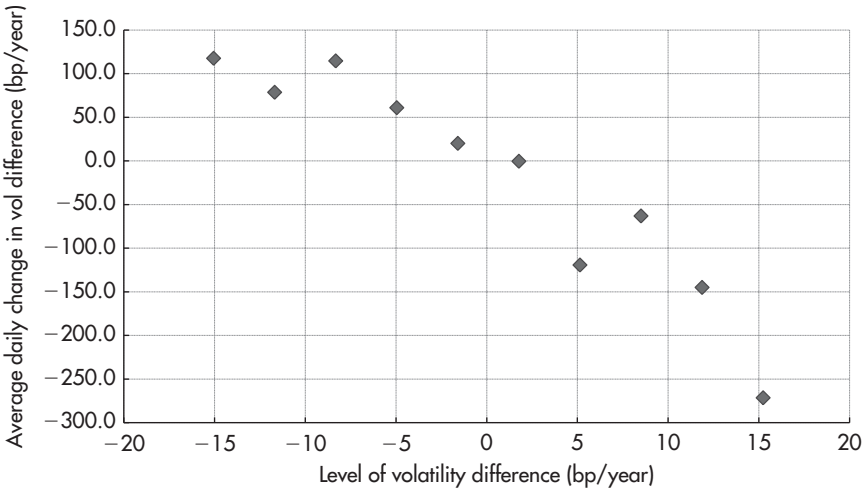


FIGURE 2.20 First-order nonparametric estimate of drift coefficient.
Source: Authors.

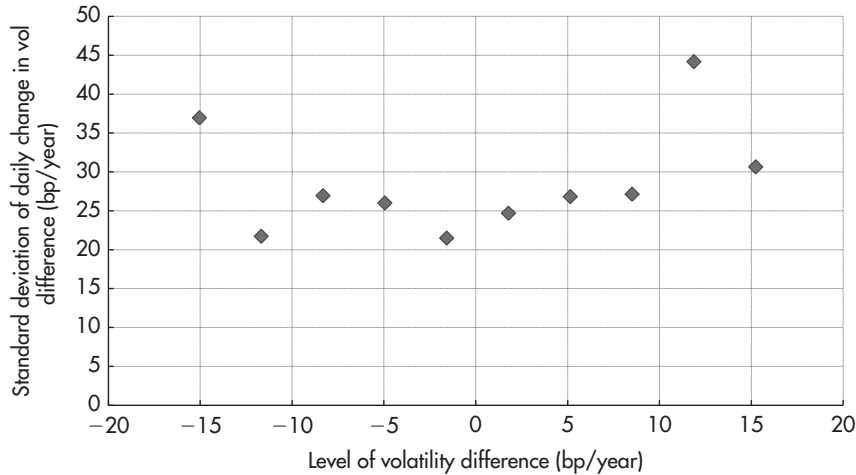


FIGURE 2.21 First-order nonparametric estimate of diffusion coefficient.
Source: Authors.

the data range seems to have exhibited greater volatility than have the other buckets. In particular, we would want to see how many observations were in each of the buckets showing elevated volatility. If each bucket contains relatively few observations, the volatility information from each bucket may not be so informative. For now, we'll use a constant to model the diffusion coefficient, but we'll note the possibility of nonlinearity when making inferences about the process.

Given the two diagnostic graphs in Figure 2.20 and Figure 2.21, we'll choose to model the volatility spread with the OU process described above. In this case, we'll need to estimate three parameters: the long-run mean, the speed of mean reversion, and the instantaneous volatility of the spread.

As the OU process has a transition density that can be represented analytically, we'll use maximum likelihood estimation to estimate the parameters of the specified process given our data. In this case, the estimate is for a long-run mean of 1.0, for an instantaneous volatility of 28.25, and for a speed of mean reversion of 16.5.

As seen from the equation above giving the conditional mean of the OU process, the expected value of the process decays exponentially, approaching the long-run mean asymptotically.

In this case, it's also useful to express the speed of mean reversion in terms of the half-life of the process, the amount of time by which the expected value of the process is halfway between the current value and the long-run mean. In our example, the half-life of the process is 15.4 calendar days, meaning we can expect the spread to close half the remaining distance toward its long-run mean every 15.4 calendar days.

Figure 2.22 shows the unconditional density (or stationary density) for the process along with the transition densities for a variety of time horizons. The current value of the spread, 12.35, is denoted by a gray vertical line.

From Figure 2.22, we see the way in which the transition densities shift their means and standard deviations over time as they approach their limit, the unconditional density for the process.

Note also that the probability of the spread being greater in the future than it is now decreases over time. For example, the probability on a one-week horizon is 18%, but the probability on a one-month horizon is only 4%. At a horizon of three months, the probability that the spread is greater than its current value of 12.35 is less than 1%, according to this model.

Another way to quantify the behavior of this volatility spread is to calculate the risk-adjusted return over various horizons. We do this by dividing the expected value of the change over the horizon by the standard deviation of the change over the same horizon. As this measure resembles the well-known Sharpe ratio, we'll refer to it by this name in the remainder of this chapter.

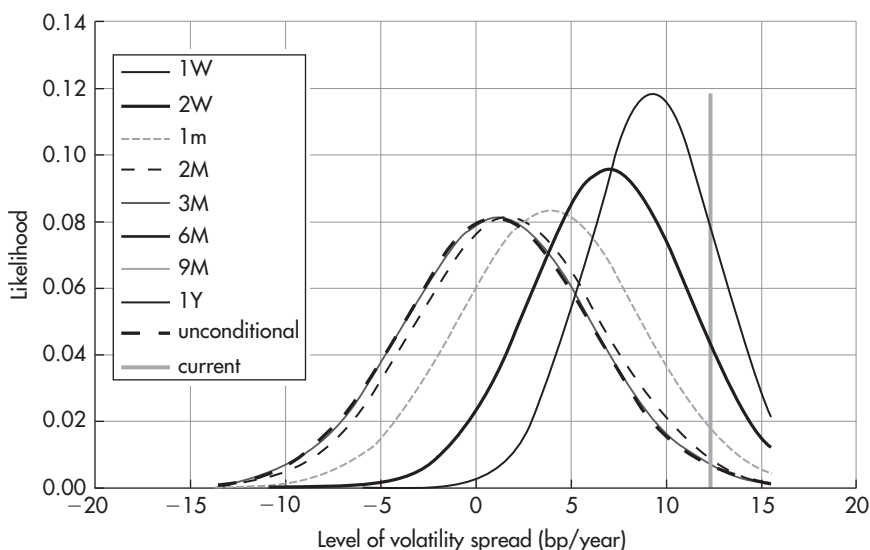


FIGURE 2.22 Unconditional and conditional densities for the volatility spread.
Source: Authors.

The Sharpe ratio on a one-week horizon is already 0.91, and this increases to 1.78 on a one-month horizon. By three months, the ratio has increased to 2.26.

However, to compare Sharpe ratios for different horizons, we should normalize all the calculations to the same time interval, since the expected values of random variables are additive, while the standard deviations increase at the rate $\sqrt{N}$, where N is the number of times the opportunity is taken. (In other words, to compare an investment with a one-year horizon to an investment with a one-month horizon, we need to consider repeating the one-month opportunity for 12 successive occasions.)

In this case, the annualized Sharpe ratio for the one-week investment horizon is 6.56, while the annualized Sharpe ratio for a one-month investment horizon is 6.12. The annualized Sharpe ratio on a three-month horizon is 4.5.

The fact that the annualized Sharpe ratios decline over time tells us that the best risk-adjusted opportunity is at short horizons. In other words, most of the performance in the trade is expected to come in the early days. The implication is that if we had repeated opportunities to commit to trades with similar risk-adjusted return profiles, it would be better for us to hold these positions for short periods of time and then reallocate our capital to other short-term opportunities.